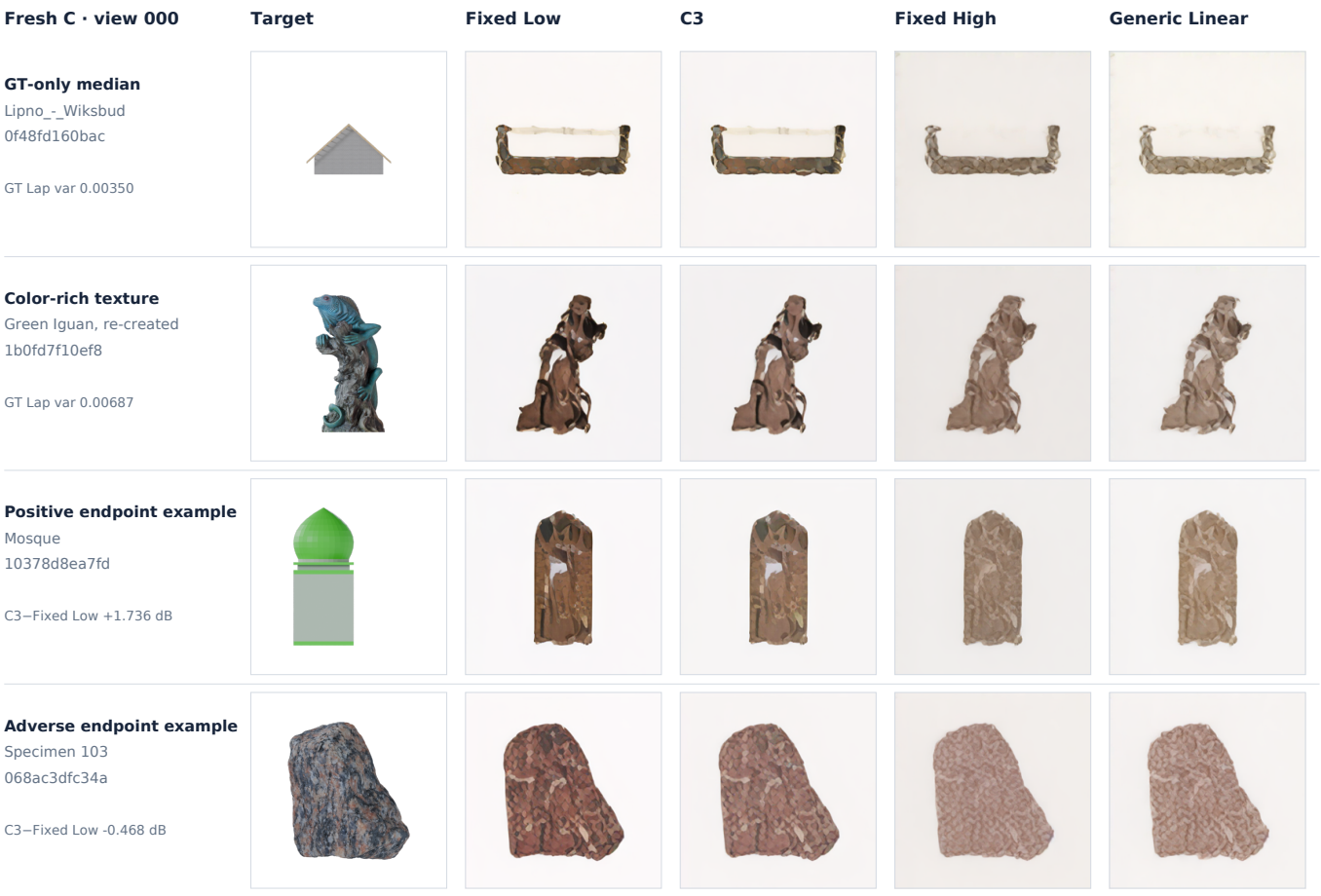


Fig. 2. Fresh C, camera 000. Every row shows Target, Fixed Low, C3, Fixed High and Generic Linear in the same full-frame view. Targets are source RGB conditioning renders, not measured ground-truth albedo. Lipno is the median target-only foreground Laplacian-variance example among the rights-cleared input-selected display set; Green Iguan adds a colorful, textured organic object selected from source properties. Mosque (+1.736 dB) and Specimen 103 (−0.468 dB) are post-outcome selected positive and adverse C3–Fixed Low FG-PSNR examples. These selected rows illustrate endpoint heterogeneity; they do not estimate prevalence or establish a cohort-wide advantage. Complete source identity and credits are in Supplementary Section S9 and the visual selection manifest.

Table 3. Selected paired color-diagnostic contrasts.

Contrast	Mean CIEDE2000 difference [95% CI]	Objects
GFL – No Adapter	−19.654 [−26.370, −13.248]	24/26 lower
LLH – GFL	+3.583 [+0.462, +6.712]	–

The 26 objects are failure-enriched, not a prevalence sample. Negative differences favor the first condition.

a ground-truth VAE encode–decode isolate had CIEDE2000 4.086 and $\Delta a^* \approx -1.35$, compared with 22.032 and +1.977 for GFL. This weakens ordinary VAE round-trip error as the sole explanation for that example, without identifying the complete cause.

LLH–GFL increased mean CIEDE2000 while reducing foreground gradient magnitude, high-frequency energy and color entropy. Lower variation therefore did not imply closer reference appearance. The denoising/runtime isolates and three failed correction candidates remain in Supplementary Section S6; they localize observations but do not validate a cause or repair.

Earlier bake results and calibration checks are reported separately in Supplementary Section S5 and are not pooled with the ten-object cohort. The GeoTex residual branch with an unset scale differs from an explicit Constant 1.0 request under the wrapper caps; we label the latter by its executed setting.

Full-object evidence and its selection. The complete views in Figure 2 and the color failure in Figure 4 use a shared full-frame display for every condition within each object; the existing color diagnostic retains its identical pixel crop. Favorable/adverse C3–GFL examples are retained in Supplementary Section S9. Outcome-selected displays illustrate behav-

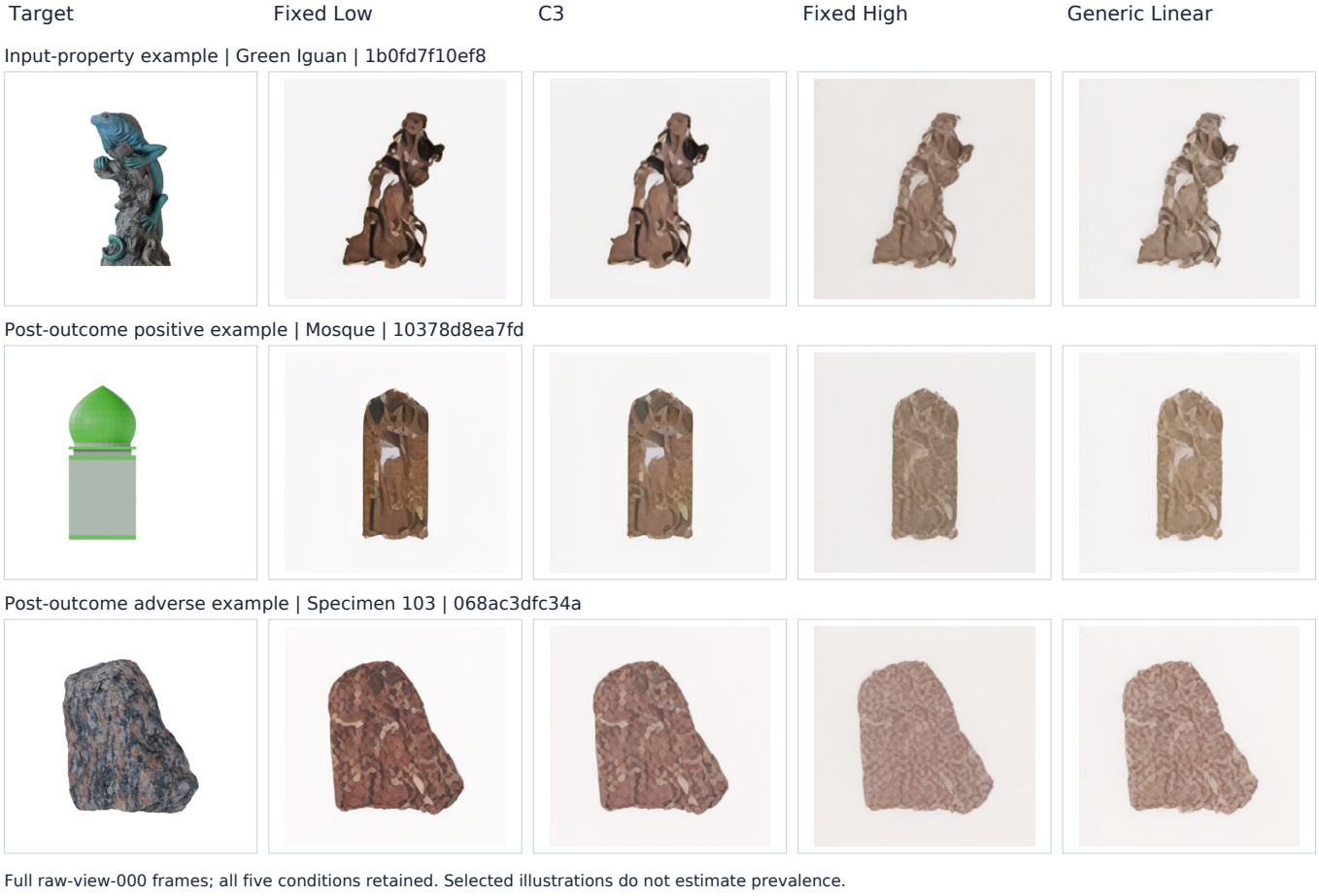


Fig. 2. Fresh C, camera 000. Every row shows Target, Fixed Low, C3, Fixed High and Generic Linear in the same full-frame view. Targets are source RGB conditioning renders, not measured ground-truth albedo. Green Iguan is a colorful, textured organic object selected from source properties. Mosque (+1.736 dB) and Specimen 103 (−0.468 dB) are post-outcome selected positive and adverse C3–Fixed Low FG-PSNR examples. These selected rows illustrate endpoint heterogeneity; they do not estimate prevalence or establish a cohort-wide advantage. Complete source identity and credits are in Supplementary Section S9 and the visual selection manifest.

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LLH–GFL increased mean CIEDE2000 while reducing foreground gradient magnitude, high-frequency energy and color entropy. Lower variation therefore did not imply closer reference appearance. The denoising/runtime isolates and three failed correction candidates remain in Supplementary Section S6; they localize observations but do not validate a cause or repair.

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Full-object evidence and its selection. The complete views in Figure 2 and the color failure in Figure 4 use a shared full-frame display for every condition within each object; the existing color diagnostic retains its identical pixel crop. Favorable/adverse C3–GFL examples are retained in Supplementary Section S9. Outcome-selected displays illustrate behavior, not prevalence. The historical texture audit does not preserve a traceable denominator or validated texture-loss labels; we therefore do not infer color-failure prevalence or texture loss from it. Variation statistics alone cannot establish fidelity.

4.4. Actual Baked Appearance: Local Benefit and External Limits

4.4.1. Native-UV protocol and primary result

We selected ten previously unbaked Fresh C objects in UID order using input availability, native UV/material compatibility, triangle count and camera-mask alignment. Only ten of the 300 screened inputs passed these criteria before the cutoff. Their generated-view outputs were known, but their baked outcomes had not been inspected during selection. This compatibility-selected pilot is not a representative sample; five earlier bake objects are reported separately.

Seven conditions share each object's source mesh, reference, seed, six projection cameras, masks, 256-pixel source resolution, atlas resolution, hole filling and eleven unseen evaluation cameras. Original geometry and UVs are retained; unseen cameras do not enter the bake. GT-256 uses the same source resolution, masks and bake parameters, with the same raw atlas coverage as each generated condition. Camera and sampling calibra-